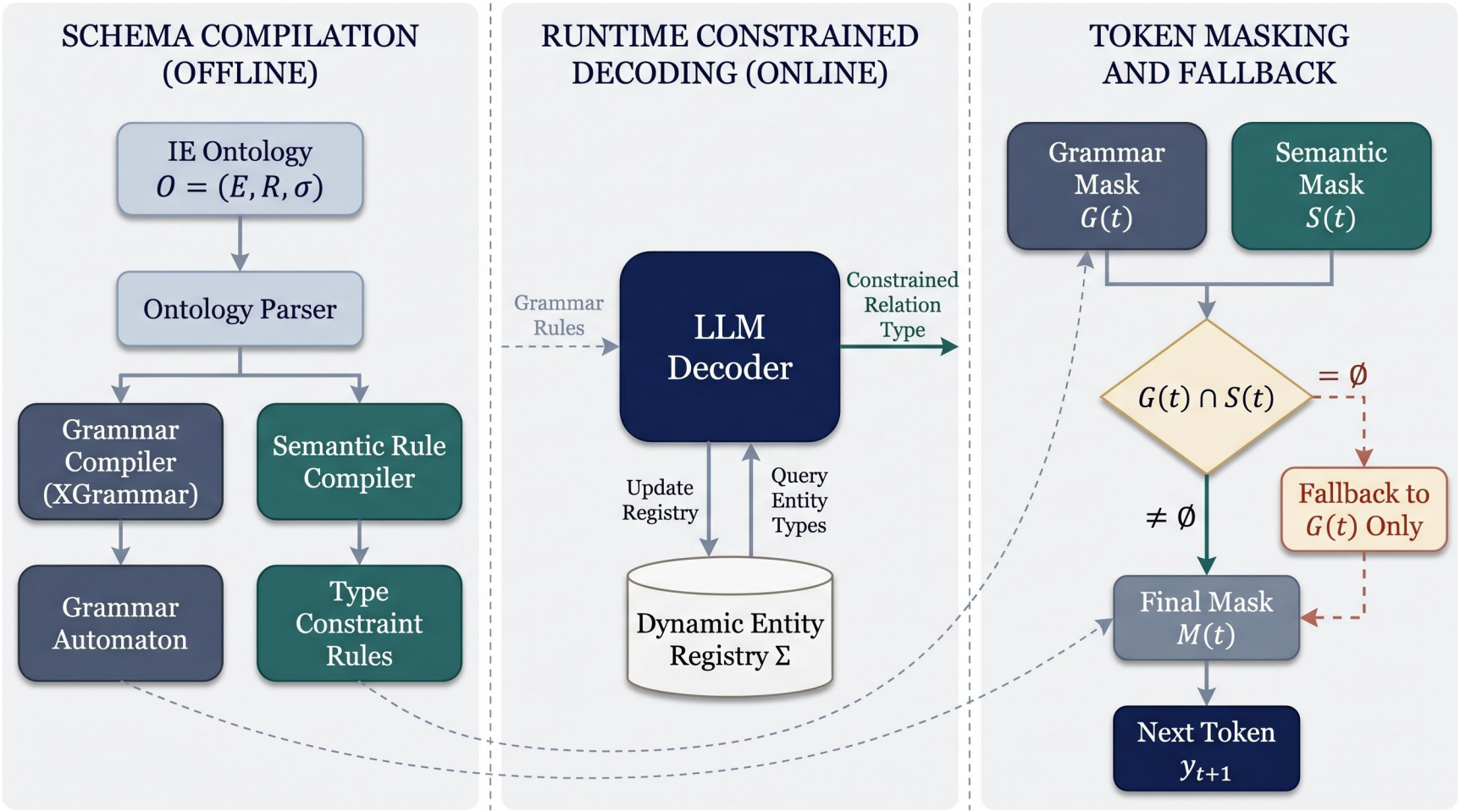

SCHEMA COMPILATION (OFFLINE)
IE Ontology $O = (E, R, \sigma)$
Ontology Parser
Grammar Compiler (XGrammar)
Semantic Rule Compiler
Grammar Automaton
Type Constraint Rules
RUNTIME CONSTRAINED DECODING (ONLINE)
Grammar Rules
LLM Decoder
Constrained Relation Type
Update Registry
Query Entity Types
Dynamic Entity Registry Σ
TOKEN MASKING AND FALLBACK
Grammar Mask $G(t)$
Semantic Mask $S(t)$
$G(t) \cap S(t)$
$= \emptyset$
$\neq \emptyset$
Fallback to $G(t)$ Only
Final Mask $M(t)$
Next Token y_{t+1}